Kigali Independent University (ULK)

Course: UML Class Diagram Modelling Assignment

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Group Members:

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REPORT 1: PROJECT DESCRIPTION REPORT

Project Title: Hotel Booking Management System

1. Business Idea:
A modern hotel needs an automated information system to manage room reservations, guest records, room availability, and billing processes efficiently.

2. Problem Statement and Purpose:
- Problem Statement: Managing hotel bookings manually using paper logs or basic spreadsheets leads to double bookings, difficulty in tracking available rooms, slow check-in/check-out processes, and billing errors.
- Purpose: The purpose of the proposed system is to centralize and automate hotel operations by tracking room statuses, managing guest reservations, and processing payments accurately.

3. Project Scope:
- Functions Covered (In-Scope):
* Managing guest profiles and contact information.
* Tracking hotel rooms, their types, prices, and availability status.
* Creating, updating, and cancelling room bookings.
* Generating and tracking booking payments.

- Functions Excluded (Out-of-Scope):
* Staff payroll and human resources management.
* Restaurant and room service food ordering.
* Developing a fully functional live software application (this project focuses strictly on designing the UML class diagram model).

4. Main Classes and Responsibilities:
- Guest: Stores personal and contact details of the customers staying at the hotel, and allows them to make or cancel bookings.
- Room: Represents individual hotel rooms, tracking their room number, category, price per night, and current availability.
- Booking: Connects a Guest to a specific Room for a defined check-in and check-out period, and calculates the total cost.
- Payment: Records financial transactions associated with a specific booking, including the payment method, amount, and status.

5. Brief Description of System Requirements:
- The system must allow guests to register and update their information.
- The system must check room availability before confirming a booking.
- The system must link each booking to one specific guest and one specific room.
- The system must record a payment transaction for every confirmed booking.